

LECTURE 9

Chapter 5 : Distributions

Key words: Distribution function, Law of a random variable, discrete random variable, continuous random variable, pmf and pdf of random variable.

In this chapter, we explore the 'quantification/ measurement' of the random variable using probability of a class of events from $\sigma(X)$. This gives us the notion of distribution function. The distribution function tell us how probability is 'distributed' among the range of the random variable.

Definition 5.1. (Distribution function)

Let $X : \Omega \rightarrow \mathbb{R}$ be a random variable on $(\Omega, \mathcal{F}, P)$. The function $F : \mathbb{R} \rightarrow \mathbb{R}$ defined by ¹

$$F(x) = P(X \leq x)$$

is called the distribution function of X .

Observe that distribution function F of X 'measures' the probability of the events $X^{-1}(-\infty, x]$ for $x \in \mathbb{R}$, i.e., for a class of events from $\sigma(X)$. Recall that the σ -field generated by the family $\{X^{-1}(-\infty, x] | x \in \mathbb{R}\}$ is $\sigma(X)$. Hence F 'describes' the probabilities of all events from $\sigma(X)$. We will see more about this later. Now we will have a closer look at the distribution function.

Theorem 0.1 *The distribution function has the following properties*

(i)

$$\lim_{x \rightarrow -\infty} F(x) = 0, \quad \lim_{x \rightarrow \infty} F(x) = 1.$$

(ii) F is non decreasing.

(iii) For $x \in \mathbb{R}$, $F(x+) = P(X \leq x)$, $F(x-) = P(X < x)$. In particular F is right continuous, with left limit exists for all x .

Proof:

(i) In the proof, we use the following.

¹we use $P(X \leq x)$ to denote $P(\{X \leq x\})$, similarly use $P(X \in B)$ to denote $P(\{X \in B\})$

For a function $g : \mathbb{R} \rightarrow \mathbb{R}$, $\lim_{x \rightarrow -\infty} g(x) = a$ ² iff whenever $x_n \downarrow -\infty$ as $n \rightarrow \infty$, we have $\lim_{n \rightarrow \infty} g(x_n) = a$.

Hence, for each sequence $\{x_n\}$ with $x_n \downarrow -\infty$, we need to compute

$$\lim_{n \rightarrow \infty} F(x_n) = \lim_{n \rightarrow \infty} P(X \leq x_n).$$

Set

$$A_n = \{X \leq x_n\}.$$

Then (exercise)

$$A_1 \supseteq A_2 \supseteq \cdots \text{ and } \bigcap_n A_n = \emptyset.$$

Therefore

$$P(A_n) \rightarrow 0.$$

i.e.,

$$P(X \leq x_n) \rightarrow 0.$$

Hence

$$\lim_{n \rightarrow \infty} F(x_n) = 0.$$

Using similar argument, we can prove (exercise)

$$\lim_{x \rightarrow \infty} F(x) = 1.$$

(ii) For $x_1 \leq x_2$, we have $\{X \leq x_1\} \subseteq \{X \leq x_2\}$ and therefore

$$P(X \leq x_1) \leq P(X \leq x_2).$$

Hence $x_1 \leq x_2 \Rightarrow F(x_1) \leq F(x_2)$.

(iii) We have to show that, for each $x \in \mathbb{R}$

$$\lim_{y \downarrow x} F(y) = P(X \leq x).$$

As in the proof of (i), it is enough to show that whenever $y_n \downarrow x$, as $n \rightarrow \infty$,

$$\lim_{n \rightarrow \infty} F(y_n) = F(x).$$

²Given a function $g : \mathbb{R} \rightarrow \mathbb{R}$, we say that $\lim_{x \rightarrow -\infty} g(x) = a$ if for each $\varepsilon > 0$, there exists a $\Delta > 0$ such that whenever $x \leq -\Delta$, we have $|g(x) - a| < \varepsilon$.

The definition of $\lim_{x \rightarrow \infty} g(x)$ and $\lim_{x \rightarrow x_0} g(x)$ are analogous. For example, $\lim_{x \rightarrow x_0} g(x) = b$ if for each $\varepsilon > 0$ there exists a $\delta > 0$ such that whenever $|x - x_0| < \delta$, we have $|g(x) - b| < \varepsilon$.

Let $y_n \downarrow x$, set

$$A_n = \{X \leq y_n\}.$$

Then

$$A_1 \supseteq A_2 \supset \cdots$$

and

$$\{X \leq x\} = \bigcap_{n=1}^{\infty} A_n. \text{ (exercise)}$$

Therefore

$$P(X \leq x) = \lim_{n \rightarrow \infty} P(A_n) = \lim_{n \rightarrow \infty} F(y_n).$$

Similarly (exercise), we can show that when ever $y_n \nearrow x$,

$$\lim_{n \rightarrow \infty} F(y_n) = P(X < x).$$

Hence $F(x-) = P(X < x)$. □

Theorem 0.2 *The set discontinuity points of a distribution function F is countable.*

Proof. Let F be the distribution function of a random variable X and

$$D = \text{set of all points of discontinuity of } F.$$

From Theorem 0.1 (iii), we get

$$D = \{x \in \mathbb{R} \mid F(x) - F(x-) > 0\}.$$

Set

$$D_n = \left\{x \in \mathbb{R} \mid F(x) - F(x-) \geq \frac{1}{n}\right\}.$$

Then

$$D = \bigcup_n^{\infty} D_n.$$

Also note that for $x \in \mathbb{R}$

$$P(X = x) = F(x) - F(x-).$$

If $x_1, x_2, \dots, x_m$ be distinct points in D_n . Then

$$\begin{aligned} 1 &\geq \sum_{k=1}^m P(X = x_k) \\ &= \sum_{k=1}^m F(x_k) - F(x_k-) \\ &\geq \frac{m}{n}. \end{aligned}$$

Therefore

$$\#D_n \leq n.$$

Therefore, D a countable union of finite sets and hence countable. $\square$

Recall that $F(x) = P(X \in (-\infty, x]), x \in \mathbb{R}$, $\mathcal{I}_1 = \{(-\infty, x] | x \in \mathbb{R}\}$ is a family of Borel sets and $\sigma(\mathcal{I}_1) = \mathcal{B}_{\mathbb{R}}$. So consider the map $B \mapsto P(X \in B)$ from $\mathcal{B}_{\mathbb{R}}$ to $[0, 1]$. Is this map defines a probability measure? Answer is affirmative as seen from the following Theorem.

Theorem 0.3 *Let X be a random variable on a probability space $(\Omega, \mathcal{F}, P)$. For $B \in \mathcal{B}_{\mathbb{R}}$, define*

$$\mu_X(B) = P(X \in B).$$

Then μ_X is a probability measure on $(\mathbb{R}, \mathcal{B}_{\mathbb{R}})$.

Proof.

$$\mu_X(\mathbb{R}) = P(X \in \mathbb{R}) = P(\Omega) = 1.$$

Also

$$\mu_X(B) = P(X \in B) \geq 0$$

for all $B \in \mathcal{B}_{\mathbb{R}}$. Let $B_1, B_2, \dots \in \mathcal{B}_{\mathbb{R}}$ be pairwise disjoint, then

$$\begin{aligned} \mu_X\left(\bigcup_{i=1}^{\infty} B_i\right) &= P\left(X \in \bigcup_{i=1}^{\infty} B_i\right) = P\left(\bigcup_{i=1}^{\infty} \{X \in B_i\}\right) \\ &= \sum_{i=1}^{\infty} P(X \in B_i) = \sum_{i=1}^{\infty} \mu_X(B_i). \end{aligned}$$

[note that B_i 's are pairwise disjoint imply that $\{X \in B_i\}$'s are pairwise disjoint] $\square$

Definition 5.2. The probability measure μ_X is called the distribution or law of the random variable X .

Note that if F is the distribution function of X , then

$$F(x) = \mu_X(-\infty, x].$$

Example 0.1 *Consider the probability space $(\Omega, \mathcal{F}, P)$ given by*

$$\Omega = \{HH, HT, TH, TT\},$$

$$\mathcal{F} = \mathcal{P}(\Omega),$$

$$P(HH) = P(HT) = P(TH) = P(TT) = \frac{1}{4}.$$

Define $X : \Omega \rightarrow \mathbb{R}$ as follows. For $\omega \in \Omega$,

$X(\omega)$ = number of H 's in ω .

Then X takes values from $\{0, 1, 2\}$. The distribution function of the random variable is given by

$$\begin{aligned} F(x) &= 0 & \text{if } x < 0 \\ &= \frac{1}{4} & \text{if } 0 \leq x < 1 \\ &= \frac{3}{4} & \text{if } 1 \leq x < 2 \\ &= 1 & \text{if } x \geq 2. \end{aligned}$$

Also, the distribution of X is

$$\mu_X(B) = \sum_{k \in B \cap \{0,1,2\}} p_k, \quad p_0 = \frac{1}{4} = p_2, p_1 = \frac{1}{2}.$$

Example 0.2 (Bernoulli distribution) Let $(\Omega, \mathcal{F}, P)$ be a probability space and $A \in \mathcal{F}$ with $p = P(A)$. Tossing a p -coin gives such a probability space with an event A . Let $X = I_A$. Here note that X takes the values 0 and 1. The distribution function of X is given by

$$\begin{aligned} F(x) &= 0 & \text{if } x < 0 \\ &= 1 - p & \text{if } 0 \leq x < 1 \\ &= 1 & \text{if } x \geq 1. \end{aligned}$$

The distribution of X is given by

$$\mu(B) = \sum_{k \in B \cap \{0,1\}} p_k, \quad B \in \mathcal{B}_{\mathbb{R}},$$

where $p_0 = 1 - p$, $p_1 = p$. The probability measure μ on $(\mathbb{R}, \mathcal{B}_{\mathbb{R}})$ are called the Bernoulli distribution and F is the corresponding distribution function. Also $X = I_A$ is an example of Bernoulli (p) random variable.

Example 0.3 (Binomial distribution with parameters (n, p)).

Let $X_1, X_2, \dots, X_n$ be n independent Bernoulli(p) random variables defined on a probability space. In fact one can define independent Bernoulli's given above through the following. Toss a p -coin n -times independently and let

$X_k = 1$ if k th toss is H and $= 0$ if the k th toss is T . Set $X = X_1 + \cdots + X_n$.
Then

$$\begin{aligned}\{X = k\} &= \{X_i = 1 \text{ for exactly } k \text{ } i\text{'s}\} \\ &= \bigcup_{\{i_1, i_2, \dots, i_k\} \subseteq \{1, 2, \dots, n\}} \{X_{i_1} = X_{i_2} = \cdots = X_{i_k} = 1, X_i = 0 \text{ otherwise}\}.\end{aligned}$$

Hence

$$\begin{aligned}P(X = k) &= \sum_{\{i_1, i_2, \dots, i_k\} \subseteq \{1, 2, \dots, n\}} P(\{X_{i_1} = X_{i_2} = \cdots = X_{i_k} = 1, X_i = 0 \text{ otherwise}\}) \\ &= \sum_{\{i_1, i_2, \dots, i_k\} \subseteq \{1, 2, \dots, n\}} p^k (1 - p)^{n-k} \\ &= \binom{n}{k} p^k (1 - p)^{n-k}.\end{aligned}$$

Hence the distribution of $X = X_1 + \cdots + X_n$ is given by

$$F(x) = \begin{cases} 0 & \text{if } x < 0 \\ \binom{n}{0} (1 - p)^n & \text{if } 0 \leq x < 1 \\ \binom{n}{0} (1 - p)^n + \binom{n}{1} p(1 - p)^{n-1} & \text{if } 1 \leq x < 2 \\ \sum_{i=0}^k \binom{n}{i} p^i (1 - p)^{n-i} & \text{if } k \leq x < k + 1, k = 2, \dots, n - 1 \\ 1 & \text{if } x \geq n. \end{cases}$$

$$\mu_X(B) = \sum_{k \in B \cap \{0, 1, \dots, n\}} \binom{n}{k} p^k (1 - p)^{n-k}.$$

The above probability measure μ_X is called the *Binomial* (n, p) distribution and F is the corresponding distribution function. A random variable with Binomial distribution as its law (distribution) is called a *Binomial random variable*.